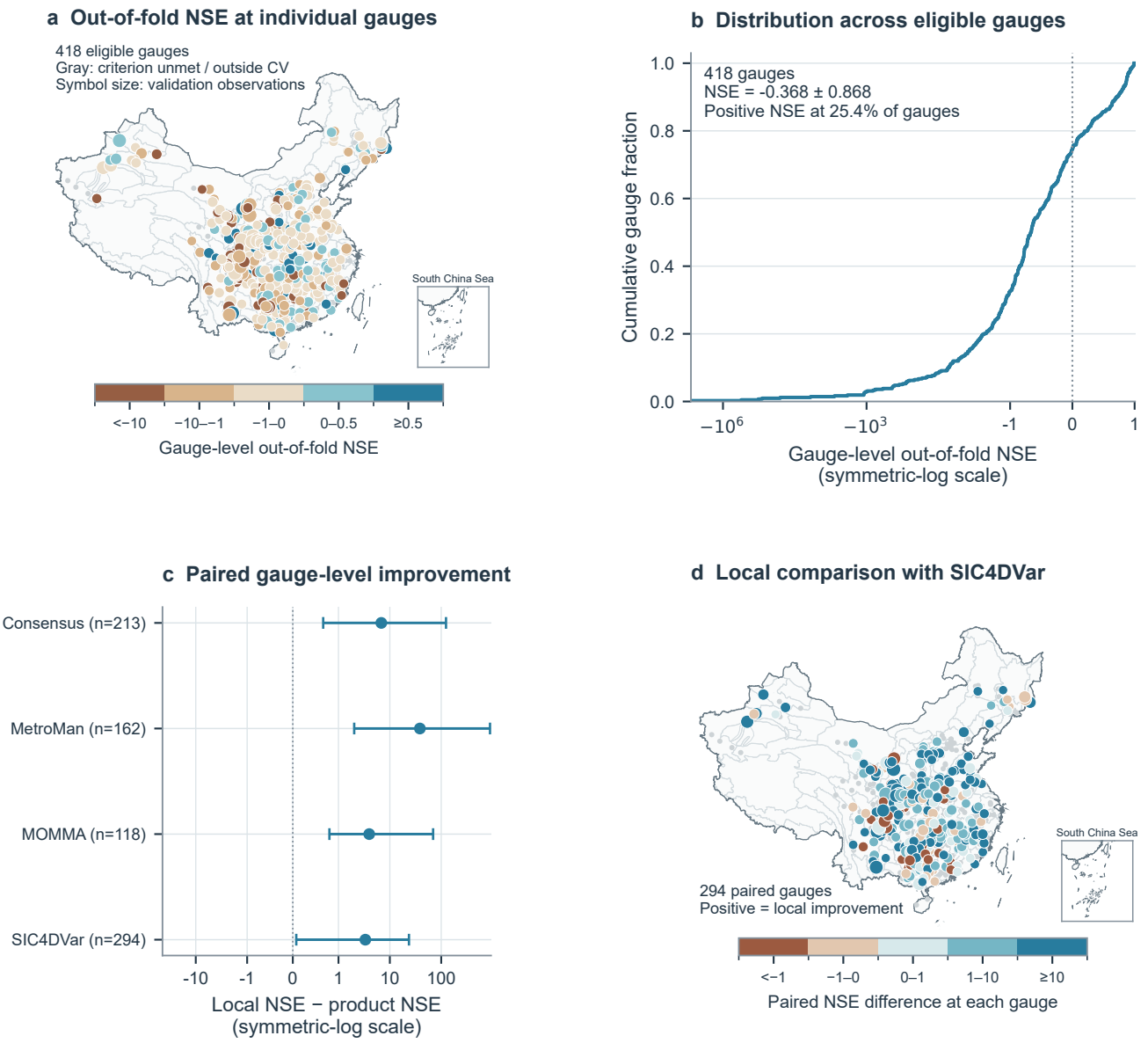

a Out-of-fold NSE at individual gauges
418 eligible gauges
Gray: criterion unmet / outside CV
Symbol size: validation observations
South China Sea
<−10
−10–−1
−1–0
0–0.5
≥0.5
Gauge-level out-of-fold NSE
b Distribution across eligible gauges
418 gauges
NSE = -0.368 ± 0.868
Positive NSE at 25.4% of gauges
Cumulative gauge fraction
1.0
0.8
0.6
0.4
0.2
0.0
−10^6
−10^3
-1
0
1
Gauge-level out-of-fold NSE
(symmetric-log scale)
c Paired gauge-level improvement
Consensus (n=213)
MetroMan (n=162)
MOMMA (n=118)
SIC4DVar (n=294)
-10
-1
0
1
10
100
Local NSE − product NSE
(symmetric-log scale)
d Local comparison with SIC4DVar
294 paired gauges
Positive = local improvement
South China Sea
<−1
−1–0
0–1
1–10
≥10
Paired NSE difference at each gauge